Factors, multiples and primes

L1-NT1 Level 1 Foundations · Number theory

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How to find and count factors, test whether a number is prime, and find HCFs and LCMs from a prime factorisation.
By the end you can list the factors of a number in pairs and test whether a number is prime; write a number as a product of primes and use it to count its factors; find HCFs and LCMs, and recognise which one a word problem needs.
Prerequisites: none. **Contests:** JMC, Junior Kangaroo, JMO, AMC 8. **Time:** about 60 minutes.

1 Lesson

The number 10 has four factors (1, 2, 5, 10) and 12 has six (1, 2, 3, 4, 6, 12). Most numbers have an even number of factors, but 36 has nine: 1, 2, 3, 4, 6, 9, 12, 18 and 36. Below 30, the numbers with an odd number of factors are 1, 4, 9, 16 and 25, all squares. We prove at the end of the unit that this always happens.

1.1 Factors in pairs

If $a \times b = n$, where a, b and n are positive whole numbers, then a and b are *factors* (or *divisors*) of n, and n is a *multiple* of a and of b. Every number n has the factors 1 and n.

Factors come in pairs: if a is a factor of n, then so is $n \div a$, and the two multiply to give n. To list the factors of n without missing any, test $1, 2, 3, \ldots$ in turn and write down each factor together with its partner. For 60 this gives

$$1 \times 60, \quad 2 \times 30, \quad 3 \times 20, \quad 4 \times 15, \quad 5 \times 12, \quad 6 \times 10.$$

Testing 7 gives nothing new, and we can stop there: if both numbers in a pair were 8 or more, their product would be at least $8 \times 8 = 64$. So 60 has 12 factors. In general the smaller number in a factor pair of n is at most $\sqrt{n}$, so you only need to test up to $\sqrt{n}$.

1.2 Primes

A *prime* is a whole number with exactly two factors, 1 and itself. The number 1 is not prime, because it has only one factor. A whole number bigger than 1 that is not prime is called *composite*. The number 2 is the only even prime, since every other even number has 2 as a factor as well as 1 and itself. The primes below 50 are worth knowing by heart:

$$2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47.$$

To test whether n is prime, it is enough to try the *primes* up to $\sqrt{n}$. Composite numbers need not be tried: if 15 divided n, then so would 3.

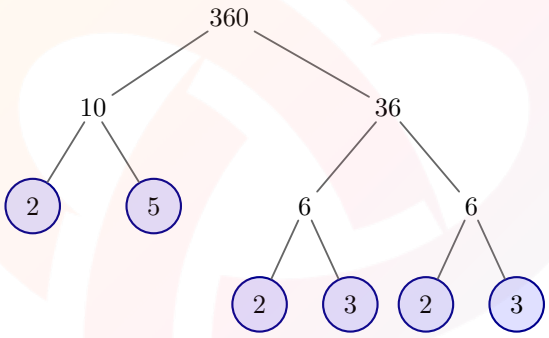
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Example 1.
Is 391 prime?

Solution.
Since $20 \times 20 = 400$ is more than 391, a factor pair of 391 other than 1×391 has its smaller number below 20. So we try the primes 2, 3, 5, 7, 11, 13, 17 and 19.
The number 391 is odd, so 2 fails. Next, $391 = 3 \times 130 + 1$, so 3 fails, and 391 does not end in 0 or 5, so 5 fails. Then $391 = 7 \times 55 + 6$, $391 = 11 \times 35 + 6$ and $391 = 13 \times 30 + 1$, so 7, 11 and 13 fail. Finally, $17 \times 23 = 391$. So 391 is not prime.
Products of two primes such as $91 = 7 \times 13$, $221 = 13 \times 17$ and $323 = 17 \times 19$ are the usual contest numbers that look prime but are not.

1.3 Prime factorisation

A composite number splits as a product of two smaller numbers, and we can keep splitting until every piece is prime. A factor tree records the splitting.



$$360 = 2 \times 2 \times 2 \times 3 \times 3 \times 5 = 2^3 \times 3^2 \times 5$$

Starting instead from $360 = 8 \times 45$ gives the same three 2s, two 3s and one 5.

Theorem 1 (prime factorisation).
Every whole number bigger than 1 can be written as a product of primes, and in only one way apart from the order of the factors.

We take the "only one way" part on trust; its proof is harder than it looks. Repeated primes are written as powers, so $360 = 2^3 \times 3^2 \times 5$. A prime to the power 0 means 1, so $2^0 = 1$.

The factors of 360 are the numbers $2^a \times 3^b \times 5^c$ where a is 0, 1, 2 or 3, b is 0, 1 or 2, and c is 0 or 1, since a factor cannot use a prime that 360 does not have, or more copies of a prime than 360 has. Different choices give different factors, so 360 has $4 \times 3 \times 2 = 24$ factors.

Theorem 2 (counting factors).
If $n = p^a \times q^b \times r^c \times \cdots$, where $p, q, r, \ldots$ are different primes, then the number of factors of n is

$$(a+1) \times (b+1) \times (c+1) \times \cdots .$$

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Example 2.
How many factors does 720 have? How many of them are odd?

Solution.
Listing pairs risks missing one, so we factorise. We have $720 = 72 \times 10 = (2^3 \times 3^2) \times (2 \times 5) = 2^4 \times 3^2 \times 5$. By Theorem 2, the number of factors is $(4+1)(2+1)(1+1) = 5 \times 3 \times 2 = 30$.
A factor $2^a \times 3^b \times 5^c$ is odd when it contains no 2, that is, when $a = 0$. That leaves 1 choice for a, 3 for b and 2 for c, so there are $1 \times 3 \times 2 = 6$ odd factors. They are 1, 3, 5, 9, 15 and 45, which confirms the count.

1.4 HCF and LCM

The *highest common factor* (HCF) of two numbers is the largest number that is a factor of both. The *lowest common multiple* (LCM) is the smallest positive number that is a multiple of both. For example, the HCF of 12 and 18 is 6 and the LCM is 36. For larger numbers we factorise. Take $84 = 2^2 \times 3 \times 7$ and $90 = 2 \times 3^2 \times 5$. A common factor may use each prime only as often as both numbers allow, so it has at most one 2 (as 90 has only one), at most one 3, and no 5 or 7. Hence the HCF is $2 \times 3 = 6$. A common multiple must contain everything in 84 and everything in 90, so it needs two 2s, two 3s, a 5 and a 7. Hence the LCM is $2^2 \times 3^2 \times 5 \times 7 = 1260$.

Theorem 3.
To find the HCF of two numbers, take each prime to the *smaller* of its two powers. To find the LCM, take each prime to the *larger* power. For two numbers a and b,

$$\text{HCF} \times \text{LCM} = a \times b.$$

The last statement holds because, for each prime, one of a and b has the smaller power and the other has the larger, so the HCF and the LCM together contain as many copies of that prime as $a \times b$ does. For example, 84 has two 2s and 90 has one; the HCF takes one 2 and the LCM takes two. Check: $6 \times 1260 = 7560 = 84 \times 90$.

Example 3.
Three lighthouses flash every 12 seconds, every 18 seconds and every 30 seconds. They all flash together at midnight. How long is it until they next all flash together?

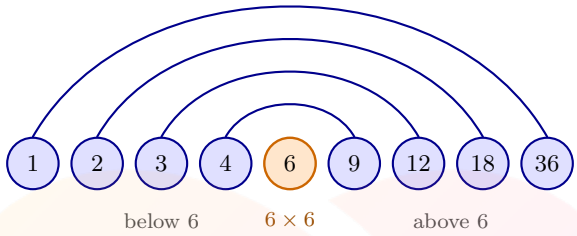
Solution.
The first lighthouse flashes at multiples of 12 seconds after midnight, and similarly for the others, so we want the LCM of 12, 18 and 30. The product $12 \times 18 \times 30 = 6480$ is a time when all three flash, but not the first one.
We have $12 = 2^2 \times 3$, $18 = 2 \times 3^2$ and $30 = 2 \times 3 \times 5$. A common multiple needs two 2s, two 3s and a 5, so the LCM is $2^2 \times 3^2 \times 5 = 180$. As a check, $180 = 12 \times 15 = 18 \times 10 = 30 \times 6$. The lighthouses next flash together after 180 seconds, which is 3 minutes.

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1.5 Square numbers and odd numbers of factors

Theorem 4.
A whole number has an odd number of factors if and only if it is a square number.

Proof. Pair each factor a of n with its partner $n \div a$. A factor is its own partner only when $a \times a = n$. If n is not a square, this never happens, so the factors split into pairs of two different numbers and there is an even number of them. If $n = m \times m$, then m is the only factor that is its own partner, so the factors are some pairs together with m on its own, an odd number.



Theorem 2 gives a second proof. Squaring doubles every power, as in $(2^3 \times 5)^2 = 2^6 \times 5^2$, so a number is a square exactly when every power in its factorisation is even. Then every bracket $(a + 1)$ is odd, and so is their product. If the number is not a square, some power is odd, its bracket is even, and the product is even. Similarly, a number is a cube exactly when every power is a multiple of 3.

Example 4.
Which whole numbers have exactly three factors? Find all of them below 100.

Solution.
Small cases first: 4 and 9 have three factors, but 16 has five. So the answer is not simply "squares".
Three is odd, so by Theorem 4 the number is a square, say $n = m \times m$. Its factors include 1, m and n, which is already three. If m had a factor d other than 1 and m, then d would be a fourth factor of n. So m has no factors except 1 and itself, and $m \neq 1$ (as 1 has only one factor), which means m is prime. Conversely, if p is prime then p^2 has $2 + 1 = 3$ factors by Theorem 2, namely 1, p and p^2.
So the numbers with exactly three factors are the squares of primes. Below 100 they are $2^2 = 4$, $3^2 = 9$, $5^2 = 25$ and $7^2 = 49$.

Common traps.

- 1 is not prime, and 2 is prime.
- Theorem 2 uses $a + 1$, not a, because a factor may leave a prime out (power 0). This is the commonest slip.
- Theorem 2 needs a *prime* factorisation. Writing $360 = 6^2 \times 10$ and computing $(2 + 1)(1 + 1) = 6$ is wrong.
- The product of two numbers is a common multiple, but usually not the lowest. A number divisible by 4 and by 6 must be divisible by 12, not necessarily by 24.

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1.6 How to spot this in a contest

1. If the words factor, divisor, multiple, prime or square appear, factorise every number in the question into primes first.
2. Events that repeat and coincide (bells, lights, buses, runners on a track) ask for an LCM. Cutting or sharing into equal pieces that are as large as possible asks for an HCF.
3. "How many factors", or "how many odd factors", means Theorem 2, restricting the choice of powers where needed.
4. "An odd number of factors", or anything about pairing factors, means square numbers.
5. Sums of primes: 2 is the only even prime, so check whether the sum is odd or even.
6. Setters like the year: $2025 = 3^4 \times 5^2 = 45^2$, $2026 = 2 \times 1013$ with 1013 prime, and 2027 is prime.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ What is the sum of all the prime numbers between 40 and 60?

Problem 2 ⋆ Explain why the sum of two primes that are both bigger than 2 can never be prime.

Problem 3 ⋆⋆ A rectangular floor measures 180 cm by 252 cm. It is to be covered with identical square tiles, without gaps, overlaps or cutting. The side length of each tile is a whole number of centimetres, and is as large as possible. How many tiles are needed?

Problem 4 ⋆⋆ A whole number N has exactly 8 factors. Two of its factors are 6 and 15. Find N.

Problem 5 ⋆⋆⋆ How many factors of 1800 are perfect squares?

Problem 6 ⋆⋆⋆ Find the smallest positive whole number n such that $18n$ is a perfect square and $12n$ is a perfect cube.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **UKMT JMC 2015 Question 11** ⋆
   Check whether a sum of three odd primes is odd or even.
2. **AMC 8 2016 Problem 7** ⋆
   Write each option as a power of a prime and look at the power.
3. **UKMT Junior Kangaroo 2019 Question 9** ⋆
   List the factor pairs of 240 in order.

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4. **UKMT JMC 2015 Question 15** ⋆⋆
   Factorise each product and compare it with the LCM of 1 to 10.
5. **UKMT Junior Kangaroo 2019 Question 13** ⋆⋆
   The edge of a cube must be a common factor of all three lengths.
6. **AMC 8 2016 Problem 20** ⋆⋆
   Decide first which values b can take, using the factors of 12 and of 15.
7. **UKMT JMC 2017 Question 23** ⋆⋆
   A number gets two marks when it is a common multiple of two of 3, 5 and 7.
8. **AMC 8 2011 Problem 24** ⋆⋆
   Is 10001 odd or even, and what does that force?
9. **UKMT JMO 2018 Question A6** ⋆⋆⋆
   Factorise 100 into primes and ask which of the two numbers can supply the 5s.
10. **AMC 8 2012 Problem 18** ⋆⋆⋆
    A composite number with no small prime factor is a product of at least two large primes.
11. **AMC 8 2020 Problem 17** ⋆⋆⋆
    Factorise 2020, then count the factors of each factor using powers.
12. **AMC 8 2018 Problem 21** ⋆⋆⋆
    Compare each remainder with its divisor, then think about common multiples.

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4 Hints

1. List the odd numbers from 41 to 59 that do not end in 5. Since $8 \times 8 = 64$, you only need to test them for the factors 3 and 7.

2. What can you say about a prime bigger than 2? What happens when you add two such numbers?

3. The tiles fit exactly along each edge, so the side length must be a factor of 180 and of 252.

4. N is a multiple of 6 and of 15, so it is a multiple of their LCM. How many factors does that LCM have already?

5. Factorise 1800 into primes. A factor is a square when every power in its factorisation is even.

6. Write $n = 2^a \times 3^b$ (other primes would only make n bigger). A square has every power even, and a cube has every power a multiple of 3.

5 Solutions

Problem 1

What is the sum of all the prime numbers between 40 and 60?

Even numbers other than 2 are not prime, and numbers ending in 5 are multiples of 5, so the candidates are 41, 43, 47, 49, 51, 53, 57 and 59. Each is less than $8 \times 8 = 64$, so a factor pair other than 1 times the number would have its smaller member at most 7. We have already ruled out 2 and 5, so it is enough to test 3 and 7.

Now $49 = 7 \times 7$, $51 = 3 \times 17$ and $57 = 3 \times 19$, so these are not prime. None of 41, 43, 47, 53 and 59 is divisible by 3 or 7, so these five are prime. Their sum is $41 + 43 + 47 + 53 + 59 = 243$.

Answer: 243

Problem 2

Explain why the sum of two primes that are both bigger than 2 can never be prime.

The only even prime is 2, so a prime bigger than 2 is odd. The sum of two odd numbers is even. (Each odd number is one more than an even number, so the sum is two more than an even number, which is even.) The two primes are each at least 3, so their sum is at least 6.

So the sum is an even number that is at least 6. It has 2 as a factor as well as 1 and itself, so it has at least three factors and is not prime.

Problem 3

A rectangular floor measures 180 cm by 252 cm. It is to be covered with identical square tiles, without gaps, overlaps or cutting. The side length of each tile is a whole number of centimetres, and is as large as possible. How many tiles are needed?

Along the 180 cm edge the tiles fit exactly, so the side length divides 180. In the same way it divides 252. The largest possible side length is therefore the HCF of 180 and 252, and with that side length the tiles do cover the floor, in a grid.

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We have $180 = 2^2 \times 3^2 \times 5$ and $252 = 2^2 \times 3^2 \times 7$. Taking the smaller power of each prime, the HCF is $2^2 \times 3^2 = 36$. So the tiles are 36 cm squares. There are $180 \div 36 = 5$ tiles along one edge and $252 \div 36 = 7$ along the other, so $5 \times 7 = 35$ tiles are needed.

Answer: 35

Problem 4

A whole number N has exactly 8 factors. Two of its factors are 6 and 15. Find N.

Since $6 = 2 \times 3$ and $15 = 3 \times 5$, any common multiple of 6 and 15 must contain a 2, a 3 and a 5. So N is a multiple of $\text{LCM}(6, 15) = 30$.

Every factor of 30 is then a factor of N. The number 30 has eight factors, namely 1, 2, 3, 5, 6, 10, 15 and 30. Since N has exactly 8 factors, these are all the factors of N. But N is a factor of itself, so N is one of these eight numbers, and so $N \leq 30$. As N is a multiple of 30, it follows that $N = 30$.

Answer: 30

Problem 5

How many factors of 1800 are perfect squares?

We have $1800 = 18 \times 100 = (2 \times 3^2) \times (2^2 \times 5^2) = 2^3 \times 3^2 \times 5^2$. A factor of 1800 has the form $2^a \times 3^b \times 5^c$ with a from 0 to 3, b from 0 to 2 and c from 0 to 2.

The factor is a perfect square when a, b and c are all even. So a is 0 or 2, b is 0 or 2, and c is 0 or 2, giving $2 \times 2 \times 2 = 8$ square factors. They are 1, 4, 9, 25, 36, 100, 225 and 900.

Answer: 8

Problem 6

Find the smallest positive whole number n such that $18n$ is a perfect square and $12n$ is a perfect cube.

If n contained a prime other than 2 or 3, that prime would appear in $18n$ and in $12n$ to the same power, which would have to be even and a multiple of 3. Removing that prime from n entirely gives a smaller number that still works. So for the smallest n we may write $n = 2^a \times 3^b$.

Then $18n = 2^{a+1} \times 3^{b+2}$ and $12n = 2^{a+2} \times 3^{b+1}$. For $18n$ to be a square, $a+1$ and $b+2$ must be even, so a is odd and b is even. For $12n$ to be a cube, $a+2$ and $b+1$ must be multiples of 3.

For a: the odd numbers are 1, 3, 5, ..., and $a+2$ is a multiple of 3 already when $a = 1$. For b: the even numbers are 0, 2, 4, ..., and $b+1$ is 1 (no), then 3 (yes) when $b = 2$. So the smallest choice is $a = 1$ and $b = 2$, which gives $n = 2 \times 3^2 = 18$.

Check: $18 \times 18 = 324 = 18^2$ and $12 \times 18 = 216 = 6^3$.

Answer: 18